
From Green Transition to Local Tensions: Critical Mineral Extraction and Fragility in Resource-Rich States

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Abstract

The rapid global expansion of clean energy technologies has intensified demand for critical minerals, specifically lithium, cobalt, nickel, copper, and rare earth elements, which are integral to renewable energy systems and battery storage. A significant proportion of these minerals are concentrated in fragile states characterised by weak governance, limited institutional capacity, and vulnerable communities.

Research Question: How does the expansion of critical mineral extraction for green technologies drive displacement and socio-political instability in fragile states?

Objective: This study examines the mechanisms through which global demand for critical minerals drives localised displacement in fragile states, and how such displacement translates into social grievance, resentment, and political instability. The study employs a structured narrative review methodology, drawing systematically on peer-reviewed literature and policy reports from the IEA, World Bank, IRENA, and OECD. Sources were selected using explicit inclusion criteria covering publications from 2014 to 2025. Thematic analysis was applied to organise findings across three analytical dimensions.

Results: Evidence from the Democratic Republic of Congo and South America's Lithium Triangle demonstrates that accelerated mineral extraction in fragile regions consistently produces physical, economic, and environmental displacement. These processes generate compounding social grievances, particularly among youth and indigenous communities, that weaken institutional trust and heighten instability.

Conclusion: Achieving an equitable green transition requires that the socio-political consequences of mineral extraction be treated as central governance challenges. Sustainable supply chain strategies must incorporate community consent, benefit-sharing mechanisms, and institutional strengthening in fragile state contexts.

Keywords: *critical minerals, green transition, displacement, fragile states, social instability, geopolitics, lithium, cobalt*

1. Introduction

The concept of the green economy, broadly defined as a pathway toward low-carbon, resource-efficient, and socially inclusive development, has gained significant global prominence in response to escalating climate change risks and the need for sustainable growth. The term was first introduced to policy discourse in 1989, when environmental economists presented Blueprint for a Green Economy to the British government [1]. Since then, governments and international institutions have increasingly committed to net-zero targets and accelerated investments in renewable energy, electric mobility, and energy storage systems as part of broader decarbonisation strategies.

Central to this transition is the rapid diffusion of green technologies, including solar photovoltaics, wind turbines, battery systems, and smart grid infrastructures, which are considerably more mineral-intensive than conventional energy systems. Global efforts to achieve net-zero goals have been largely shaped by international frameworks such as the Paris Agreement, which, unlike the earlier Kyoto Protocol, recognised climate change as a shared responsibility and called on all countries to set binding emissions targets [2]. These commitments have stimulated unprecedented expansion in electric vehicles, offshore wind capacity, and large-scale battery storage systems, each requiring substantial volumes of specific critical minerals.

However, beneath this global environmental imperative lies a deepening structural tension. According to the International Energy Agency (IEA) and the World Bank, demand for minerals such as lithium, cobalt, nickel, copper, and rare earth elements is projected to increase dramatically in the coming decades [3, 4]. Many of these resources are geographically concentrated in a small number of regions, notably the Democratic Republic of Congo (DRC) for cobalt, and the Lithium Triangle of Chile, Argentina, and Bolivia. Critically, several of these mineral-rich regions are also classified as fragile states, characterised by weak governance structures, limited regulatory capacity, and communities highly vulnerable to the disruptions associated with large-scale resource extraction.

This convergence between surging global mineral demand and the fragility of key supply regions raises important questions that the existing literature has addressed unevenly. While the geopolitical dimensions of mineral competition have received growing scholarly attention, the localised socio-political consequences of extraction, particularly displacement and its downstream effects on instability, remain underanalysed. Scholars such as Sovacool et al. (2020) have highlighted how the energy transition reshapes geopolitical rivalries through mineral supply chains [5], while Terminski (2014) has documented the systemic nature of mining-induced displacement [9]. However, the specific

intersection of these dynamics within fragile state governance contexts remains a gap in the literature that this study addresses.

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The remainder of this paper is structured as follows. Section 2 presents the methodology. Section 3 reviews existing literature on critical mineral extraction, green economy transitions, and displacement. Section 4 examines the political economy of critical mineral extraction. Section 5 analyses the mechanisms through which extraction causes displacement, drawing on case evidence from the DRC and the Lithium Triangle. Section 6 traces the pathway from displacement to resentment and instability. Section 7 addresses geopolitical implications. Section 8 presents conclusions and policy implications.

2. Literature review

2.1 The Green Economy and Critical Mineral Demand

The global transition toward low-carbon energy systems has generated substantial scholarly interest in the role of critical minerals as enablers and constraints of decarbonisation. The IEA (2021) estimates that a sustainable development scenario would require a quadrupling of mineral inputs by 2040, with lithium, cobalt, nickel, and rare earth elements among the most demand-intensive [3]. The World Bank (2020) similarly projects demand increases of up to 500 percent for several key minerals by 2050 [4]. These projections have elevated critical minerals from an industrial procurement concern to a central issue of energy security and geopolitical strategy.

Research by the International Renewable Energy Agency (IRENA, 2023) situates this demand surge within broader shifts in the global political economy, arguing that the energy transition is transferring strategic importance from fossil fuel exporters toward mineral-rich states, many of which lack the institutional capacity to manage extraction responsibly [6]. This structural observation is foundational to the present study's concern with fragile states.

2.2 Geopolitical Competition for Critical Minerals

Scholars increasingly argue that control over critical mineral supply chains has become a central element of contemporary geopolitical competition. Sovacool et al. (2020) contend that the energy transition is likely to produce new forms of rivalry centred on mineral supply chains, noting that unlike fossil fuels, these chains involve complex transnational networks of extraction, refining, manufacturing, and recycling that create multiple points of strategic vulnerability [5]. The

concentration of processing capacity in China, which dominates the refining of rare earth elements and battery materials, has prompted industrialised economies to pursue supply chain diversification through domestic investment, strategic partnerships, and trade agreements.

Governments including those of the United States, European Union member states, Japan, and India have adopted national critical mineral strategies aimed at securing long-term access to supply through overseas investment and diplomatic engagement. This securitisation of mineral supply has intensified extraction pressure on resource-rich developing states, often with insufficient attention to the governance and community consequences of accelerated mining.

2.3 Mining-Induced Displacement

The literature on mining-induced displacement documents its systemic and multidimensional character. Terminski (2014) classifies mining displacement into physical, economic, and cultural categories, arguing that conventional humanitarian frameworks are inadequate to capture the full range of losses experienced by affected communities [9]. Garcia-Zavala et al. (2025) provide recent quantitative evidence of the non-green effects of lithium extraction in South America, demonstrating measurable environmental and economic harm to local communities in the Atacama region [8].

Amnesty International (2023) has documented forced evictions and violations of Free, Prior, and Informed Consent (FPIC) requirements in the DRC's Kolwezi cobalt mining district, providing specific evidence of how industrial extraction operates in the absence of effective governance [7]. These studies collectively establish that extraction-driven displacement is not incidental but structurally produced by the intersection of resource demand and governance failure.

2.4 Displacement, Fragility, and Instability

The literature connecting displacement to political instability in fragile states draws on a broader body of work on forced migration and governance. Myers et al. (2024) present evidence from multiple country contexts showing that forced displacement weakens social cohesion and strains host community relations when governance responses are inadequate [10]. The OECD (2025) frames state fragility as both a cause and a consequence of displacement, noting that weak institutions reduce the capacity to manage extraction-related disruptions while also making communities more vulnerable to their effects [11].

Research on youth radicalisation in displacement contexts, summarised by Taylor and Kaplan (2023) and UNDP (2023), identifies blocked economic opportunity, unresolved grievance, and loss of

belonging as structural drivers of vulnerability to extremist recruitment, with direct relevance to mining-affected youth populations in fragile states [16, 17]. Taken together, this literature establishes the conceptual foundation for the present study's argument that extraction-driven displacement in fragile states generates instability through identifiable and addressable mechanisms.

2.5 Research Gap

Despite the growing bodies of literature on critical mineral geopolitics, mining-induced displacement, and fragile state governance, these strands have rarely been integrated into a unified analytical framework. Studies of mineral supply chains tend to focus on interstate competition and energy security, while displacement research focuses on humanitarian consequences without engaging with the green economy context now driving extraction expansion. This study addresses that gap by tracing the full pathway from green economy demand to mineral extraction, displacement, and instability within fragile state contexts.

3. Methodology

3.1 Research Design

This study employs a structured narrative review methodology to examine the socio-political consequences of critical mineral extraction in fragile states. This approach combines systematic source selection with interpretive analytical synthesis, enabling the study to draw on evidence from multiple disciplines, including political science, environmental economics, development studies, and international relations, while maintaining analytical coherence around the central research question.

3.2 Justification of Methodological Choice

A structured narrative review was selected in preference to alternative methodological approaches for three reasons. First, a full systematic review with PRISMA protocol was not appropriate for this study. The intersection of mineral extraction, community displacement, and fragile state governance is inherently interdisciplinary, spanning literatures that use different terminologies, methodologies, and quality standards. The standardised appraisal tools designed for systematic reviews, developed primarily for clinical and single-discipline research, are poorly suited to synthesising evidence across this range. Moreover, the exhaustive database coverage required by a strict PRISMA protocol was not feasible within the scope of this study.

Second, a scoping review, while appropriate for mapping emerging fields, does not permit the degree of analytical synthesis and theoretical interpretation that this study's research question requires.

Specifically, tracing the causal pathway from extraction to displacement to instability demands interpretive engagement with evidence, not merely its mapping.

Third, a primary empirical approach involving fieldwork, such as interviews, ethnographic observation, or community surveys, was beyond the scope of this study, which aims to synthesise and critically analyse existing evidence rather than generate new community-level data. Primary fieldwork in fragile state extraction zones also raises significant ethical and logistical challenges that would require a substantially different research design. The structured narrative review is therefore the most epistemologically appropriate design for this study's objectives.

3.3 Search Strategy and Source Selection

Literature was sourced from three academic databases, namely Scopus, Web of Science, and Google Scholar, supplemented by policy repositories from the International Energy Agency (IEA), the World Bank, IRENA, OECD, UNHCR, and Amnesty International. Boolean search strings combined terms across four conceptual clusters: (1) critical minerals, lithium, cobalt, rare earth elements; (2) green economy, energy transition, decarbonisation; (3) displacement, forced relocation, community resettlement; and (4) fragile states, state fragility, conflict-affected regions. Publications from 2014 to 2025 were included, with the lower boundary reflecting the post-Paris Agreement acceleration of green technology deployment. Foundational theoretical works published before 2014 were included where directly relevant.

3.4 Inclusion and Exclusion Criteria

Table 1: Inclusion and Exclusion Criteria

Criteria	Inclusion	Exclusion
Publication type	Peer-reviewed articles, IEA/World Bank/OECD/UNHCR reports, Amnesty International investigations	Opinion pieces, blogs, non-indexed grey literature
Topic relevance	Directly addresses mineral extraction, community displacement, or fragile state governance	Mining studies with no social or political dimension

Geography	Global, with focus on fragile or developing states	Studies limited exclusively to high-income, stable economies
Time period	2014 to 2025 (foundational works pre-2014 permitted)	Sources without clear relevance to contemporary green transition
Language	English	Non-English sources

3.5 Thematic Analysis

Following source selection, a thematic analysis approach was applied to organise and interpret the evidence. Sources were read systematically and coded according to recurring patterns relevant to the research question. Codes were inductively derived from the literature and deductively informed by the study's three analytical dimensions: the political economy of critical mineral extraction; the mechanisms through which extraction produces physical, economic, and environmental displacement; and the pathway through which displacement generates social grievance and political instability in fragile state contexts.

Themes were reviewed iteratively across sources, refined for internal consistency, and used to structure the findings presented in Sections 5, 6, and 7. Divergent findings and counter-evidence were explicitly incorporated into the analysis to ensure that the narrative synthesis reflects the full range of evidence rather than selectively assembled support for a predetermined conclusion.

3.6 Limitations

This study acknowledges six limitations. First, reliance on secondary sources means that community-level lived experiences are mediated through institutional and academic accounts that may not fully capture local realities or the diversity of community responses to extraction. Second, the exclusive use of English-language sources introduces a linguistic bias, as relevant scholarship in French, Spanish, Swahili, and Portuguese may offer perspectives not represented in this review. Third, heavy reliance on multilateral institutional reports may systematically underrepresent community grievances or frame extraction harms within development frameworks that favour continued investment. Fourth, in-depth case evidence is drawn primarily from two contexts, the DRC and the Lithium Triangle, which may not capture the full diversity of extraction-displacement dynamics in other fragile states such as Guinea, Zimbabwe, or Indonesia. Fifth, the rapidly evolving policy landscape means some findings

relating to export controls and supply chain agreements may be partially superseded. Sixth, the study design does not permit definitive causal inference; the relationships identified are analytical linkages supported by converging evidence, not experimentally established causal mechanisms.

3.7 Ethical Considerations

As a literature review involving no primary data collection, this study did not require formal ethical approval from an institutional review board and no human participants were involved. However, the ethical dimensions of conducting research on vulnerable communities in fragile states warrant explicit consideration. The communities at the centre of this analysis, including mining-affected populations in the DRC, indigenous Atacamenno people in Chile, and artisanal miners in fragile states, are among the most structurally disadvantaged groups in the global economy.

This study is mindful of the risk of representing affected communities as passive victims of structural forces rather than as agents with their own forms of resistance, adaptation, and political organisation. The authors have endeavoured to centre the documented experiences of affected communities and to avoid framings that subordinate local welfare to global supply chain efficiency. The analysis of displacement-related grievance and instability is presented as evidence of governance failure and structural injustice, not as evidence of inherent instability in specific populations or regions.

4. Political economy of critical mineral extraction

4.1 Structural Concentration and Strategic Dependencies

Critical minerals are those deemed essential to a nation's economic growth, technological capability, and national security. The rapid scaling of clean energy technologies has transformed mineral supply chains into a strategic pillar of the green economy, creating intense competition among states and corporations for access to finite and unevenly distributed resources. The spatial distribution of these minerals is highly concentrated: the DRC holds a majority of global cobalt reserves, Chile and Australia dominate lithium supply, and China controls the processing and refining of many rare earth elements and battery materials [3]. This structural geography creates asymmetric dependencies that shape both international relations and the conditions faced by communities living in extraction zones.

The global market value of key energy transition minerals, including copper, lithium, nickel, cobalt, graphite, and rare earth elements, has already reached approximately USD 325 billion and is projected to exceed USD 770 billion by 2040. These projections have triggered a global race for resource access,

reshaping investment flows, diplomatic partnerships, and governance priorities in mineral-rich regions.

4.2 Fragile States and Value Chain Positioning

Fragile states are structurally positioned at the lowest tier of the critical mineral value chain: raw extraction with minimal processing, refining, or manufacturing capacity. This means they absorb the full social and environmental costs of extraction while capturing a disproportionately small share of the economic value generated. Without domestic refining capability, fiscal benefit-sharing frameworks, or robust regulatory institutions, fragile mineral-rich states risk functioning primarily as resource exporters for industrialised economies rather than as beneficiaries of the green transition they are enabling.

Table 2 presents the 30 critical minerals identified by the Government of India in 2023, illustrating the breadth of strategic applications across sectors including renewable energy, defence, electronics, and advanced manufacturing [21].

Table 2: Critical Minerals and their Strategic Applications (Government of India, 2023)

S.No	Critical Mineral	Key Applications
1	Antimony	Flame retardants, lead acid batteries
2	Beryllium	Telecommunications, aerospace components
3	Bismuth	Low melting alloys, cosmetics
4	Cadmium	Nickel-cadmium batteries, solar energy, nuclear technology
5	Cobalt	High performance batteries, superalloys
6	Copper	Electrical components, renewable energy infrastructure
7	Gallium	Compound semiconductors, LEDs
8	Germanium	Fibre optics, infrared optics
9	Graphite	Battery anodes, lubricants

S.No	Critical Mineral	Key Applications
10	Hafnium	Nuclear control rods, semiconductor chips
11	Indium	Touch screens, LCD panels
12	Lithium	EV batteries, grid storage
13	Molybdenum	Metallurgy, renewable energy
14	Niobium	Superconducting magnets, high-strength steel
15	Nickel	Lithium-ion batteries
16	PGE	Catalysts
17	Phosphorus	Fertiliser, iron-phosphate batteries
18	Potash	Agriculture
19	Rare Earth Elements	Defence guided systems, medical imaging
20	Rhenium	Jet engine superalloys
21	Selenium	Electronics, metallurgy
22	Silicon	Solar PV panels, semiconductor chips
23	Strontium	LCD displays
24	Tantalum	High performance capacitors, electronics
25	Tellurium	Thin film solar cells
26	Tin	Electronics, coatings
27	Titanium	Aerospace alloys, medical implants

S.No	Critical Mineral	Key Applications
28	Tungsten	Hard metals, defence equipment, cutting tools
29	Vanadium	Grid scale redox flow batteries
30	Zircon	Ceramics, nuclear industry

4.3 Counter Arguments: Does Extraction Always Harm?

Before examining the displacement mechanisms in detail, it is important to acknowledge that the relationship between mineral extraction and community welfare is not uniformly negative. A balanced assessment requires engagement with the evidence for positive outcomes as well as harms.

First, mining revenues have, in some contexts, funded meaningful public investment. In Botswana, diamond revenues were channelled through state institutions into education, healthcare, and infrastructure, producing one of the most successful development trajectories in sub-Saharan Africa. This demonstrates that resource extraction is not inherently harmful, as the governance framework through which revenues are managed is the critical variable.

Second, formal and informal mining employment provides livelihoods in regions where few economic alternatives exist. In artisanal mining communities in the DRC and Zimbabwe, mining, despite its dangers, represents the primary source of cash income for hundreds of thousands of families. Abrupt closure or exclusion of artisanal miners from industrial expansion zones therefore carries its own displacement costs that must be weighed against the harms of extraction.

Third, some benefit-sharing agreements between mining companies, governments, and communities have delivered genuine local development. Community development funds, royalty-sharing mechanisms, and local procurement requirements, where enforced, have produced roads, schools, and water systems in otherwise neglected areas.

Fourth, state-led extraction models in Bolivia and Chile have attempted to retain a greater share of mineral value domestically than pure foreign direct investment models, using revenues to cross-subsidise social programmes.

Fifth, growing investment in mineral recycling and circular economy approaches may reduce future extraction pressure in fragile states, as secondary supply becomes more economically competitive with primary mining.

These counter-examples do not invalidate the central argument of this paper; they clarify it. The harms documented in this study are not an inevitable consequence of extraction per se, but of extraction conducted in the absence of strong governance, genuine community consent, and equitable benefit-sharing. The displacement paradox is therefore, at its core, a governance failure that is acutely likely in fragile states, but not determined by the mere presence of minerals.

5. Findings: How Mineral Extraction Causes Displacement

5.1 The Displacement Paradox

While critical minerals are essential for decarbonisation, their extraction creates what scholars have termed a displacement paradox: the pursuit of global environmental stability frequently triggers localised socio-economic collapse [9]. In fragile states with weak regulatory frameworks and limited community protections, this paradox is most acute. Displacement driven by mineral extraction is not merely a residential disruption; it is a systemic dispossession that operates through multiple, overlapping mechanisms.

5.2 Extraction Methods as Drivers of Displacement

The physical footprint of a mine and the method used to extract ore are primary determinants of the scale and type of community displacement. Surface and open-pit mining, which is the dominant method used in cobalt extraction in the DRC, requires the wholesale removal of topsoil, vegetation, and infrastructure. This produces immediate physical displacement, as entire settlements are razed to accommodate excavation pits, often without adequate notice, compensation, or legal process. Amnesty International documented multiple forced evictions at industrial cobalt and copper mines in the Kolwezi region of the DRC, where communities were removed without the Free, Prior, and Informed Consent (FPIC) required under international law [7].

Brine extraction for lithium, which is the predominant method used in Chile and Argentina's Lithium Triangle, operates differently but causes equally serious disruption. This method pumps lithium-rich brine from underground aquifers to evaporation ponds on the surface, consuming extraordinary volumes of freshwater. Research by Garcia-Zavala et al. (2025) demonstrates that lithium extraction in the Atacama region has measurably reduced local water availability, threatening the agricultural and

pastoral livelihoods of indigenous Atacameno communities [8]. This constitutes slow-onset environmental displacement: people are not physically removed, but the conditions sustaining their way of life are progressively destroyed.

5.3 Case Study: Cobalt Extraction in the DRC

The DRC provides the starkest documented example of extraction-driven displacement in a fragile state. The country accounts for over 70 percent of global cobalt production, a mineral essential for lithium-ion batteries used in electric vehicles and consumer electronics. The Kolwezi mining district in Lualaba Province has experienced industrial expansion at a pace that has outstripped any meaningful governance response.

Communities in Kolwezi have faced repeated forced evictions to make way for open-pit expansion. Amnesty International's 2023 investigation found that residents were relocated without genuine consultation, offered inadequate or no compensation, and resettled in areas with inferior access to farmland and services [7]. Artisanal miners, many of them children, work in dangerous conditions in the gaps between industrial operations, reflecting the absence of functional labour regulation. The result is a layered displacement: physical relocation, economic dispossession, and the erosion of social structures within communities that previously relied on subsistence agriculture and small-scale mining.

The DRC's fragile state status, evidenced by its consistently low ranking on the Fragile States Index, means that grievances generated by extraction have limited peaceful institutional outlets. Land disputes cannot be resolved through dysfunctional courts; compensation claims cannot be enforced against politically connected mining companies; and affected communities have no meaningful access to environmental regulation. This institutional vacuum is precisely the condition under which extraction-generated grievance translates into broader instability.

5.4 Case Study: Lithium Extraction in South America's Lithium Triangle

The Lithium Triangle, spanning the arid highlands of Chile, Argentina, and Bolivia, holds approximately 58 percent of the world's identified lithium reserves. The region has become the focal point of a global scramble for battery-grade lithium, with mining operations expanding rapidly to meet surging electric vehicle demand.

In Chile's Atacama Desert, lithium extraction has generated significant conflict with the indigenous Atacameno (Lickanantay) people, who have inhabited the region for millennia. The primary mechanism of harm is hydrological: brine extraction depletes the underground aquifer system that

sustains the wetland ecosystems on which Atacameno pastoralism and agriculture depend. Research confirms that water tables in extraction zones have declined measurably, causing wetland shrinkage and degrading the grazing land on which llama and alpaca herding is based [8]. The Atacameno communities face cultural displacement, which is a permanent loss of the ecological conditions that define their identity, livelihood, and territorial relationship.

In Bolivia, lithium extraction ambitions have generated a different form of tension. The Bolivian government's state-led model has periodically conflicted with the expectations of local communities who anticipated employment and development benefits but experienced limited tangible gains. This gap between anticipated benefit and experienced reality is a recurring pattern in resource extraction in fragile and developing states, and is a significant independent driver of resentment.

5.5 A Taxonomy of Displacement

Drawing on these cases and the broader literature, displacement driven by mineral extraction can be understood through three analytically distinct but empirically overlapping categories:

- **Physical Displacement:** the forced relocation of households and communities from land required for mining operations, processing facilities, or infrastructure. This is the most visible form and the most directly governed, or ungoverned, by legal frameworks such as FPIC.
- **Economic Displacement:** the severance of communities from productive land, water, or other resources on which their livelihoods depend, even when residents are not physically relocated. Fencing of ancestral grazing land or contamination of agricultural soils constitutes economic displacement that forces migration without formal eviction.
- **Environmental Displacement:** mining-induced degradation of water, soil, or air quality that renders land uninhabitable or unproductive over time, effectively producing environmental refugees. This category is particularly associated with brine extraction, acid mine drainage, and tailings contamination.

6. Findings: From Displacement to Resentment and Instability

6.1 State Fragility as a Structural Amplifier

Forced displacement is frequently treated as a humanitarian emergency requiring immediate relief. While these responses are necessary, they are insufficient for understanding the longer-term socio-political consequences of extraction-driven displacement in fragile states. For many affected

communities, displacement is not a temporary interruption followed by recovery; it is a prolonged condition shaped by insecurity, institutional failure, economic exclusion, and social dislocation [18, 20].

One of the clearest structural conditions amplifying displacement-to-instability pathways is state fragility. People do not lose trust in institutions only because violence erupts around them; they also lose trust because the systems that should protect them, including courts, regulators, ombudsmen, and police, cease to function in their interests. Where states are unable to provide security, regulate disputes, deliver services, or enforce compensation, daily life becomes progressively unviable [11, 12]. This dynamic is self-reinforcing: extraction revenues may accrue to central governments or political elites rather than affected communities, deepening the perception that the state serves extractive interests over citizen welfare and increasing the likelihood that grievance will be expressed through protest or conflict rather than institutional channels.

6.2 Psychological and Social Dimensions

The damage caused by displacement is not only material; it is also deeply psychological, and this dimension is consistently underweighted in extraction governance frameworks. People who are forced to move commonly experience loss on multiple levels simultaneously, including home, income, education, routine, family networks, and a sense of control over the future. Trauma reduces trust, narrows future planning horizons, and increases reluctance to invest in livelihoods or social relationships [15, 20]. In that sense, psychological harm is not separate from economic recovery; it is often one of the reasons recovery remains fragile.

6.3 Youth, Exclusion, and Radicalisation Risk

Young people face particular risks in displacement contexts. The combination of interrupted schooling, weakened family structures, and prolonged periods without meaningful employment or civic engagement creates conditions in which radicalisation risk increases. This vulnerability is not primarily ideological; it is structural, rooted in blocked futures, unresolved grievance, and the search for belonging and recognition [16, 17]. In mining-affected communities where youth observe that resource wealth bypasses their families while generating environmental harm, this grievance takes on an explicitly political character.

6.4 Host Communities and Social Cohesion

Large inflows of displaced persons can create real pressure on housing, public services, employment markets, and natural resources in receiving areas. Tensions are especially likely where host populations already experience poverty and perceive support as inequitably distributed. However, hostility is not inevitable. Evidence from multiple country contexts suggests that where displaced populations are integrated into local service systems and economic participation is enabled, social cohesion is more likely to hold [10, 19].

6.5 Lessons from Post-Conflict Recovery

The experience of Sierra Leone following its civil war offers instructive parallels for extraction-affected communities. After the conflict, some returning refugees were supported through entrepreneurship initiatives that enabled them to restart economic life. The durability of some of those initiatives demonstrates that even modest, well-targeted support can produce lasting results when displaced people are treated as economic actors rather than permanent aid recipients. Rebuilding income alone is insufficient when trauma and social stigma remain unaddressed [14]. This lesson applies directly to extraction-affected communities in fragile states, where economic restitution must be accompanied by psychological and social support to be durable.

7. Geopolitical Implications

The global transition toward clean energy technologies has fundamentally elevated the strategic importance of critical minerals, transforming what were previously niche industrial inputs into key determinants of national energy security and technological competitiveness. As a result, access to these minerals has become a central axis of contemporary geopolitical competition.

The scale of projected demand growth is extraordinary. According to the IEA, demand for key minerals used in clean energy technologies is expected to nearly triple by 2030 and quadruple by 2040 in scenarios aligned with global climate goals [3]. Lithium demand alone rose by approximately 30 percent in 2023, driven primarily by electric vehicle expansion. Demand for nickel, cobalt, graphite, and rare earth elements has also grown substantially, with annual increases ranging between 8 and 15 percent in recent years.

A critical geopolitical vulnerability is the concentration of both extraction and processing in a small number of countries. Cobalt production is heavily concentrated in the DRC. Lithium reserves are dominated by the Lithium Triangle. China controls the refining and processing capacity for many critical minerals, including rare earths and battery-grade materials, a position it has used strategically

in trade disputes. Reports indicate that more than half of strategic minerals are now subject to some form of export control, reflecting intensifying geopolitical competition.

Sovacool et al. (2020) argue that the energy transition is producing new forms of geopolitical rivalry centred on mineral supply chains that differ structurally from fossil fuel competition. Unlike oil, which flows through relatively visible pipelines and tanker routes, critical mineral supply chains involve complex transnational networks of extraction, refining, component manufacturing, and recycling, creating multiple points of vulnerability [5]. IRENA's analysis further notes that this transition is reshaping geopolitical alignments by shifting strategic focus from fossil fuel reserves toward mineral endowments and processing capabilities [6].

Countries including the United States, members of the European Union, Japan, and India have responded to supply chain vulnerabilities by investing in domestic exploration, recycling technologies, and strategic sourcing partnerships. These initiatives reflect a broader recognition that mineral supply security has become inseparable from energy security and national defence. However, they also risk intensifying extraction pressure on fragile states by increasing the premium placed on rapid resource access over responsible governance of extraction.

8. Conclusion

This study set out to examine how the expansion of critical mineral extraction for green technologies drives displacement and socio-political instability in fragile states. Through a structured narrative review of peer-reviewed literature and policy evidence, the paper has traced a pathway from global green economy demand to mineral extraction, localised displacement, and governance-mediated instability.

The evidence from the DRC and the Lithium Triangle confirms that large-scale mineral extraction in fragile state contexts systematically produces physical, economic, and environmental displacement through mechanisms including forced eviction, hydrological depletion, and livelihood severing. In fragile states, where institutional frameworks to manage these disruptions are absent or captured, displacement generates deep grievances that translate into reduced institutional trust, youth radicalisation risk, host community tensions, and broader political instability.

Importantly, this paper has also shown that extraction-driven harm is not inevitable. The displacement paradox is a governance failure, not a geological one. Where strong governance frameworks, genuine community consent, equitable benefit-sharing, and robust environmental regulation are in place,

extraction can coexist with community welfare and support development. This distinction points toward institutional reform and regulatory strengthening as the primary levers of change.

Three policy implications follow from the findings. First, international frameworks governing critical mineral supply chains must incorporate enforceable community protection standards, including genuine FPIC processes, transparent benefit-sharing mechanisms, and independent environmental monitoring, as baseline requirements for investment rather than optional additions. Second, fragile states hosting mineral extraction require targeted institutional strengthening support alongside investment flows, so that governments possess the capacity to regulate extraction in the public interest. Third, the psychological and social dimensions of displacement, including trauma, loss of cultural identity, and youth exclusion, must be treated as integral components of remediation and resettlement planning rather than secondary concerns to be addressed after economic compensation.

The green transition cannot be sustained on foundations of social injustice. Addressing the displacement paradox is therefore not merely an ethical obligation; it is a strategic necessity for the long-term viability of the global clean energy transition. Future research should examine community-level outcomes through primary fieldwork in fragile state extraction zones, assess the effectiveness of specific benefit-sharing and FPIC frameworks, and investigate the conditions under which extraction governance reforms have succeeded in producing equitable outcomes.

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